Test Cases

The assignment requires manual, dry-run, invalid-input, duplicate-record, empty-record and boundary testing.

Test Case	Input / Operation	Expected Output	Result
TC01	Add new student	Student added successfully	Pass
TC02	Add duplicate roll number	Student already exists	Pass
TC03	Search existing roll	Student details displayed	Pass
TC04	Search non-existing roll	Student not found	Pass
TC05	Update student name	Record updated	Pass
TC06	Invalid update option	Invalid choice	Pass
TC07	Delete existing roll	Record deleted	Pass
TC08	Delete non-existing roll	Student not found	Pass
TC09	Calculate average	Correct average displayed	Pass
TC10	Find highest scorer	Highest scorer displayed	Pass
TC11	Search by department	Matching students displayed	Pass
TC12	Count students	Correct count displayed	Pass
TC13	Display when empty	No records available	Pass
TC14	Unique departments	Set displayed	Pass
TC15	Unique subjects	Set displayed	Pass